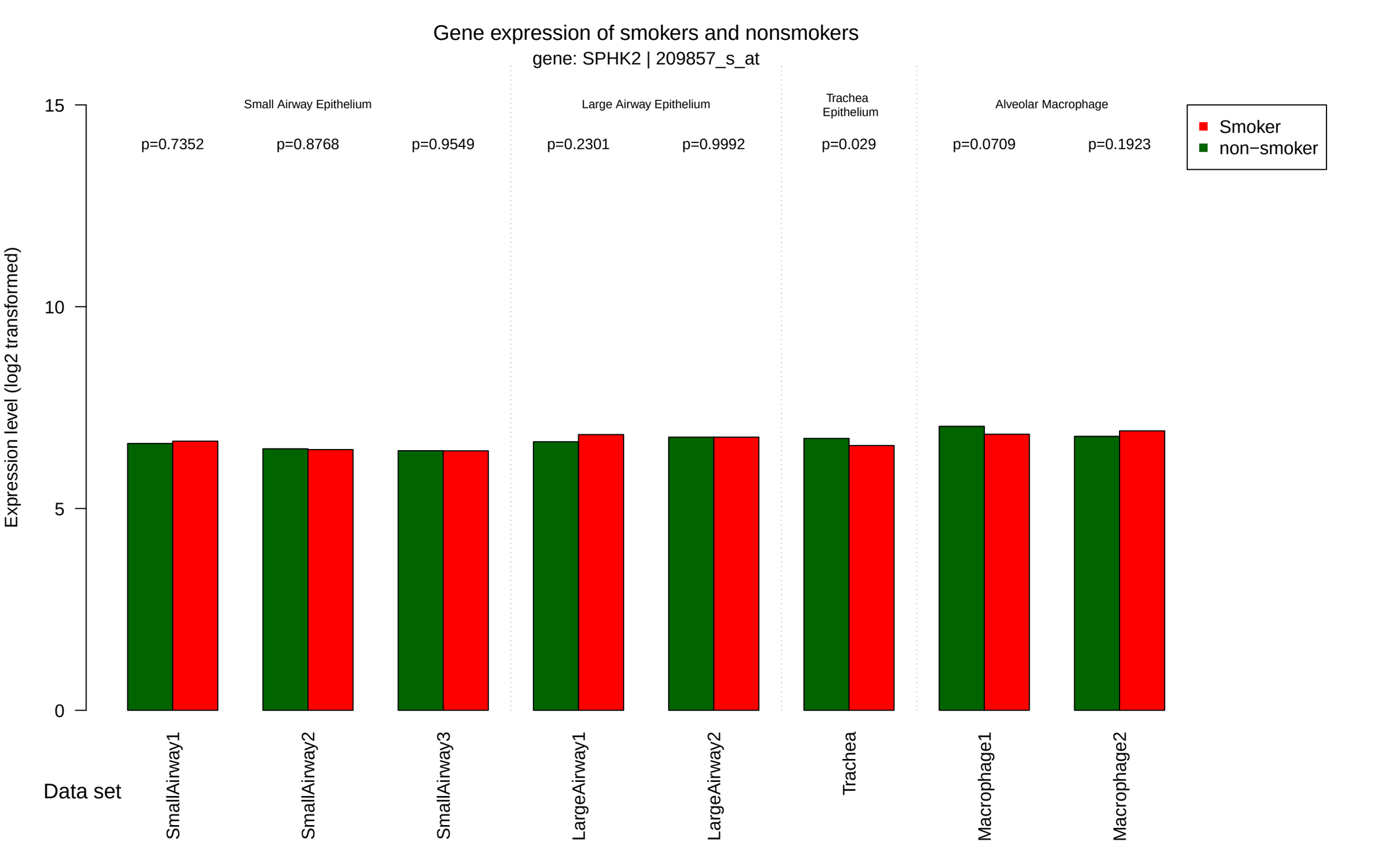
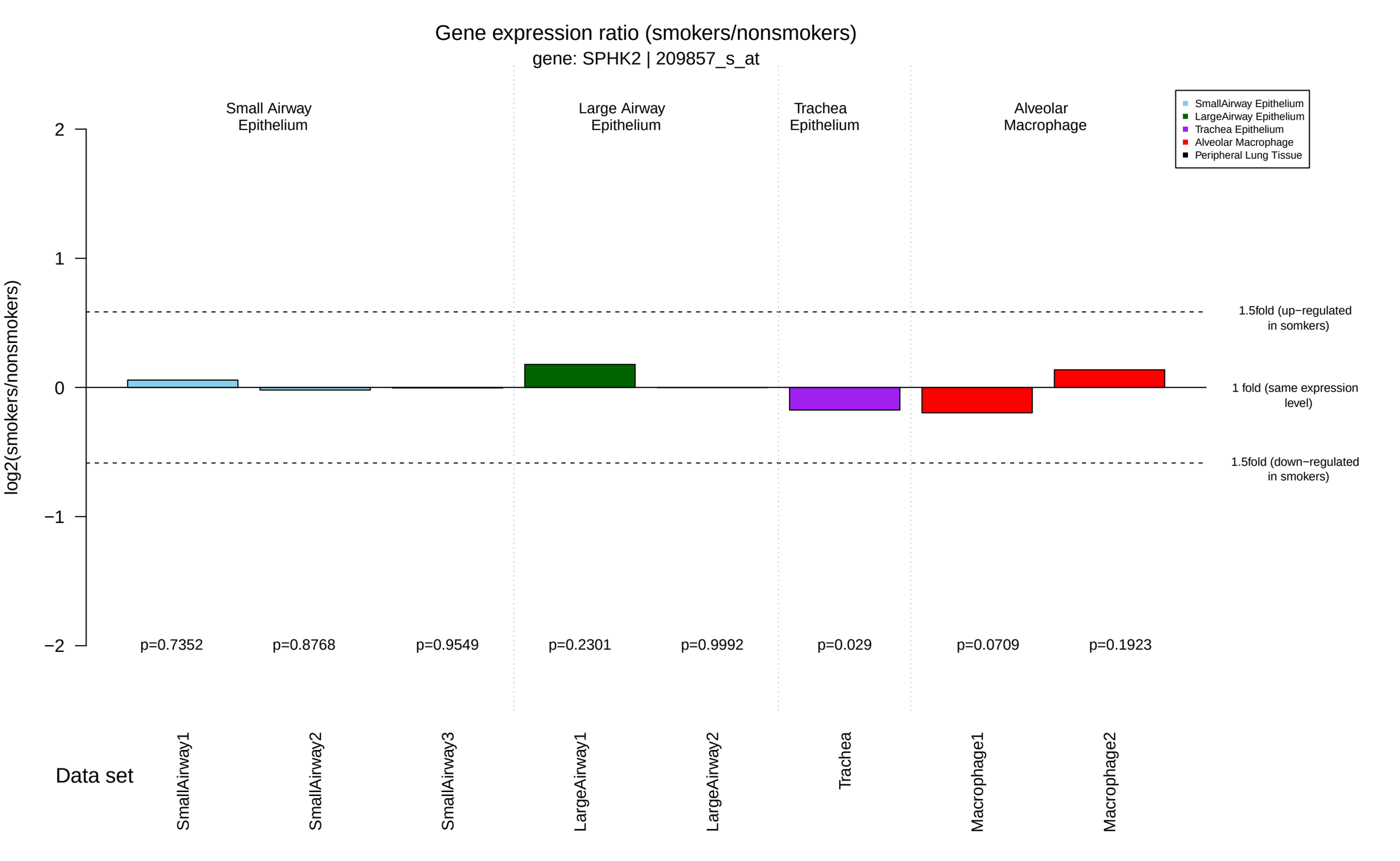
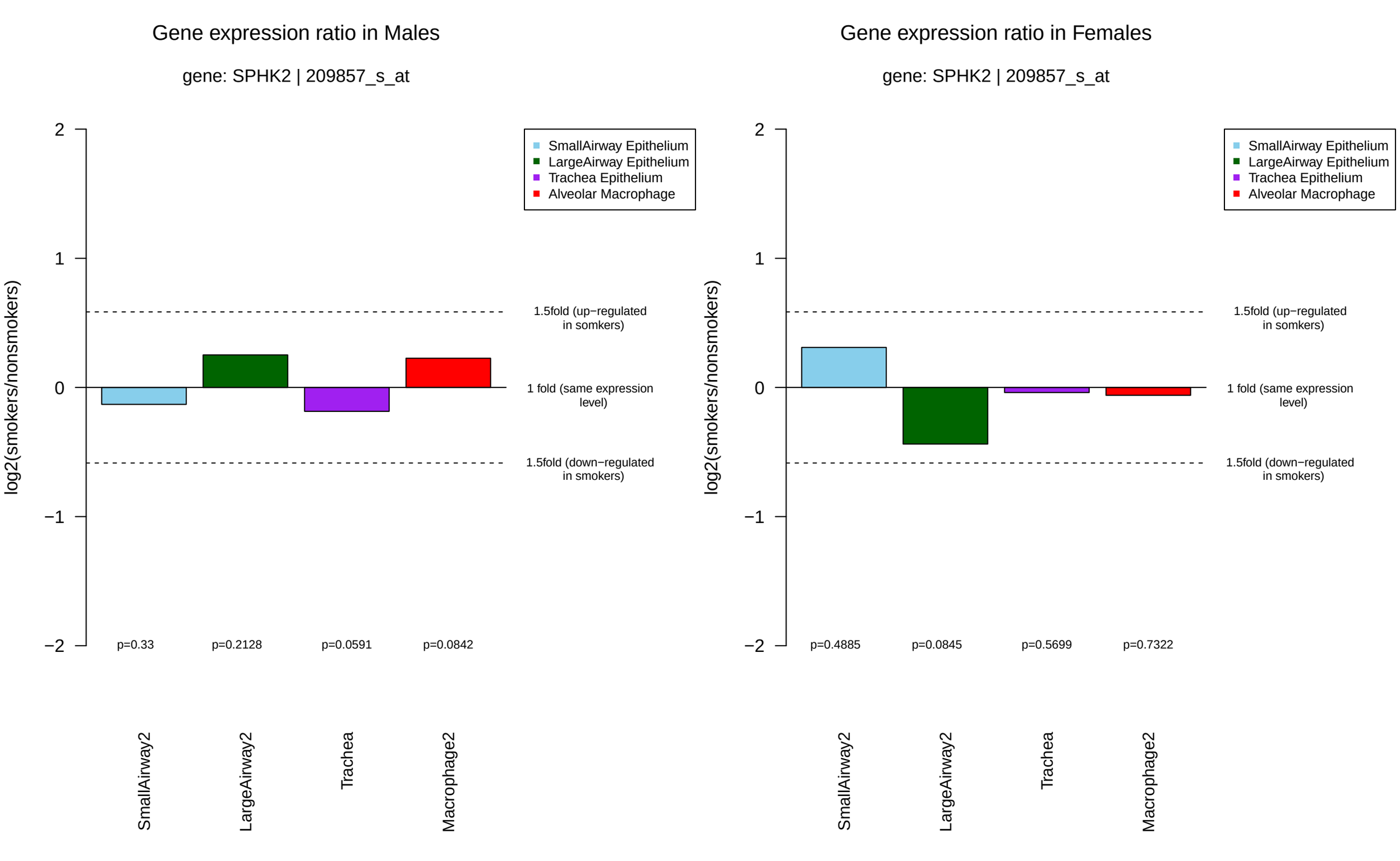
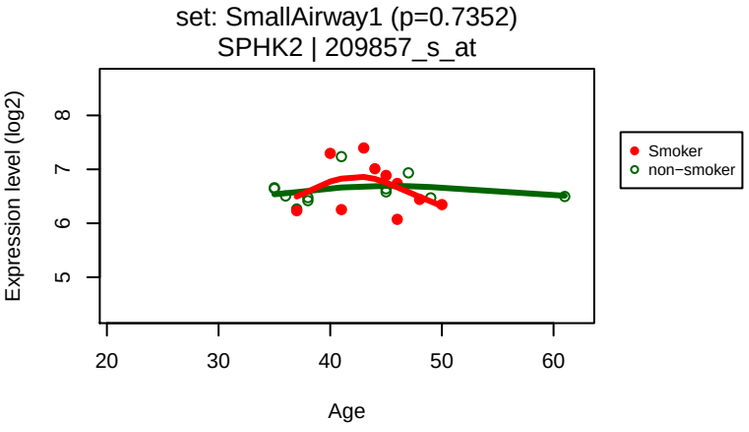
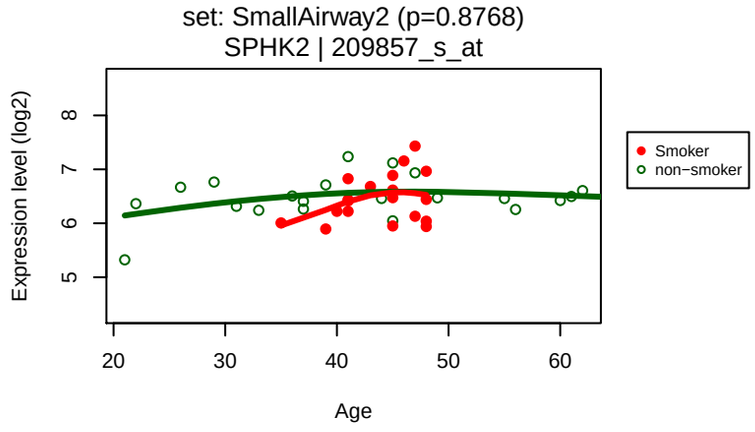
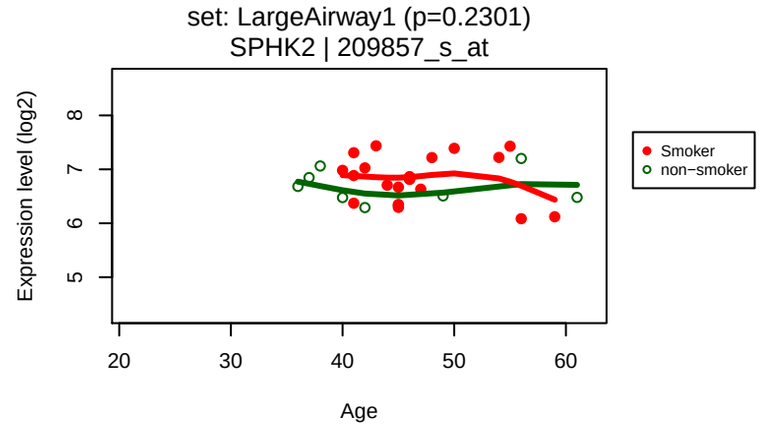

Gene expression of smokers and nonsmokers
gene: SPHK2 | 209857_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.7352
p=0.8768
p=0.9549
p=0.2301
p=0.9992
p=0.029
p=0.0709
p=0.1923
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

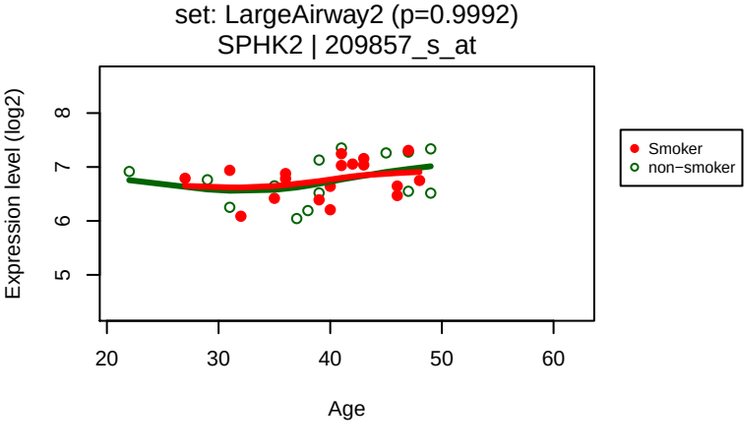

Gene expression ratio (smokers/nonsmokers)
gene: SPHK2 | 209857_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7352
p=0.8768
p=0.9549
p=0.2301
p=0.9992
p=0.029
p=0.0709
p=0.1923
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

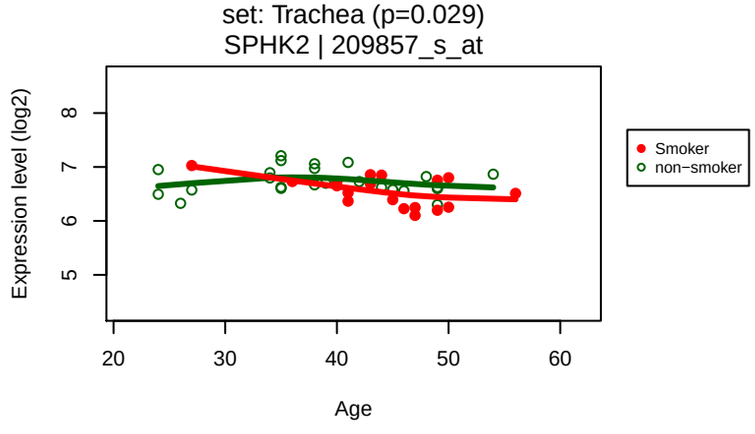

Gene expression ratio in Males
gene: SPHK2 | 209857_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.33
p=0.2128
p=0.0591
p=0.0842
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: SPHK2 | 209857_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4885
p=0.0845
p=0.5699
p=0.7322
SmallAirway2
LargeAirway2
Trachea
Macrophage2

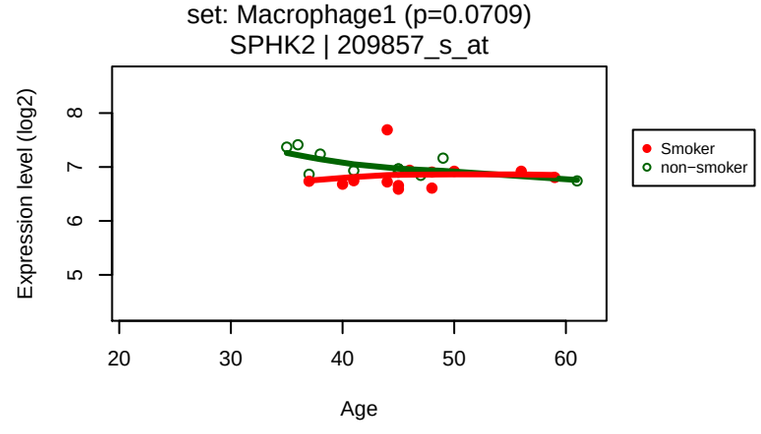

set: SmallAirway1 (p=0.7352)
SPHK2 | 209857_s_at
Expression level (log2)
Age
Smoker
non−smoker

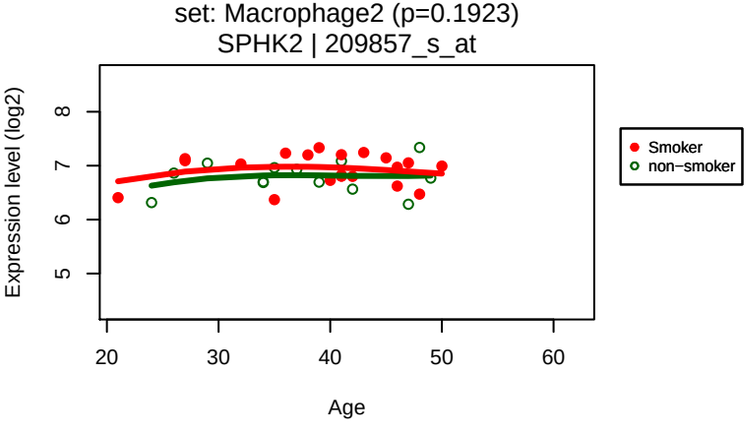

set: SmallAirway2 (p=0.8768)
SPHK2 | 209857_s_at
Expression level (log2)
Age
Smoker
non−smoker

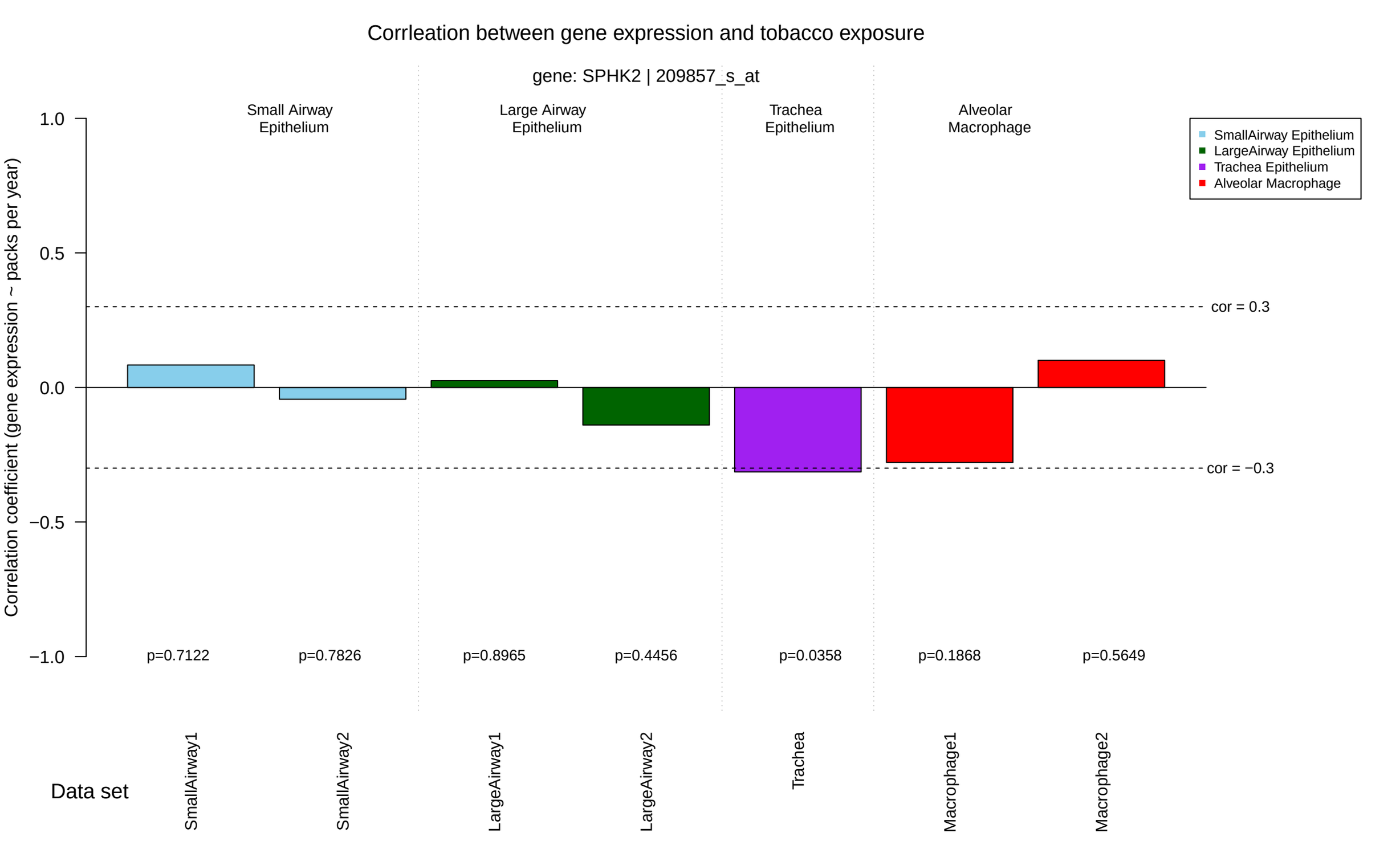

set: LargeAirway1 (p=0.2301)
SPHK2 | 209857_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.9992)
SPHK2 | 209857_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.029)
SPHK2 | 209857_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0709)
SPHK2 | 209857_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.1923)
SPHK2 | 209857_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: SPHK2 | 209857_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.7122
p=0.7826
p=0.8965
p=0.4456
p=0.0358
p=0.1868
p=0.5649
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2